

PROGRAM – 3

Code a Dockerized Node.js Application

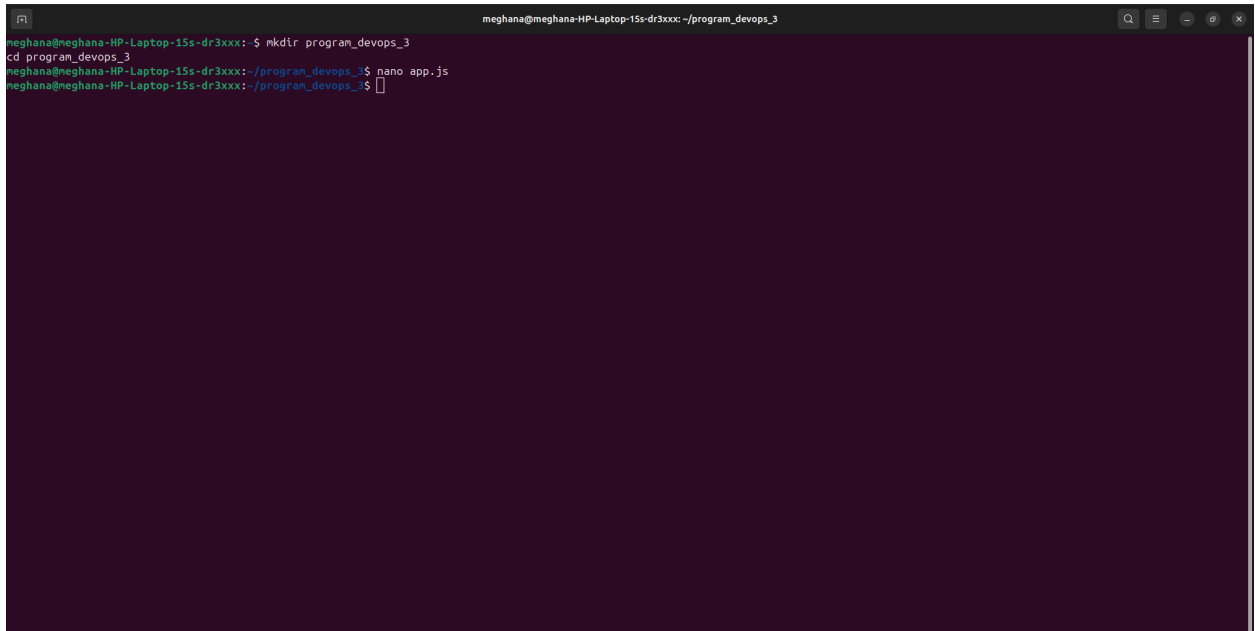
Project Name: program_devops_3

Technology Used: Node.js, Docker

Project Structure

program_devops_3/

- |
- |— app.js
- |— package.json
- |— Dockerfile



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3
meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program_devops_3
cd program_devops_3
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3 $ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3 $
```

Step 1 – Node.js Application (app.js)

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3
GNU nano 7.2 app.js
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Hello from Dockerized Node.js App - program_devops_3");
});

app.get("/health", (req, res) => {
  res.json({ status: "UP" });
});

const PORT = 4000;

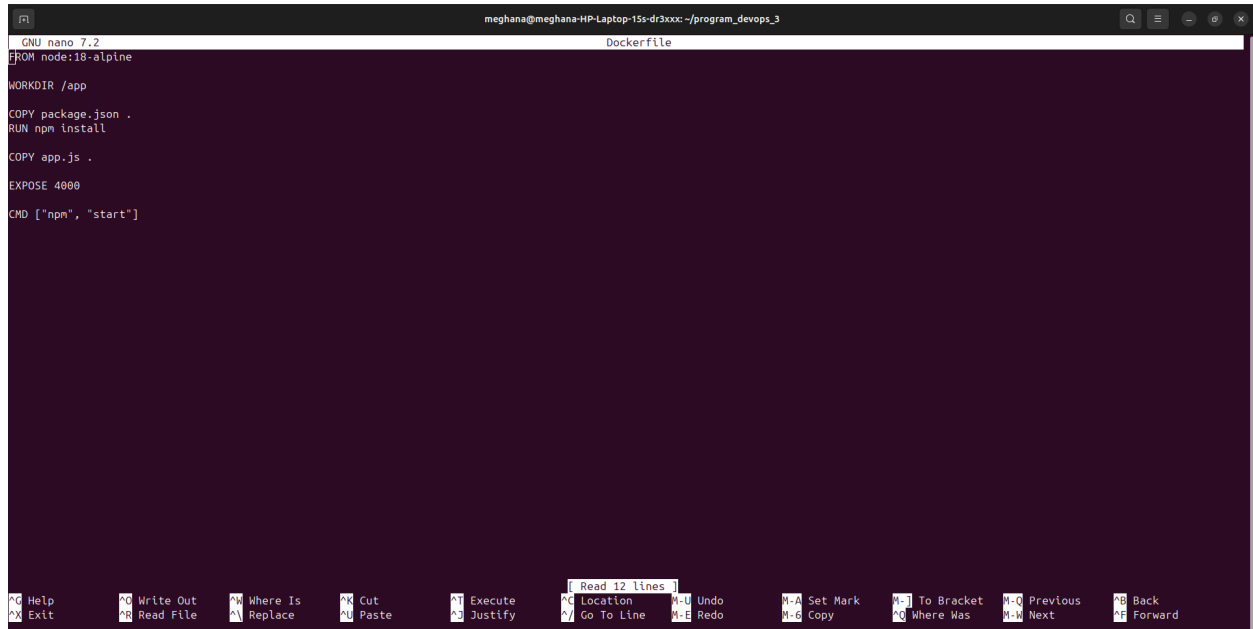
app.listen(PORT, "0.0.0.0", () => {
  console.log(`Server running on port ${PORT}`);
});

[ Read 16 lines ]
^O Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^A Set Mark   ^J To Bracket ^O Previous  ^B Back
^X Exit      ^R Read File  ^M Replace    ^N Paste      ^D Justify    ^V Go To Line ^E Redo       ^G Copy       ^G Where Was  ^N Next      ^F Forward
```

Step 2 – Dependency File (package.json)

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3
GNU nano 7.2 package.json *
{
  "name": "program_devops_3",
  "version": "1.0.0",
  "main": "app.js",
  "scripts": {
    "start": "node app.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

Step 3 – Dockerfile



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3
GNU nano 7.2 Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY package.json .
RUN npm install

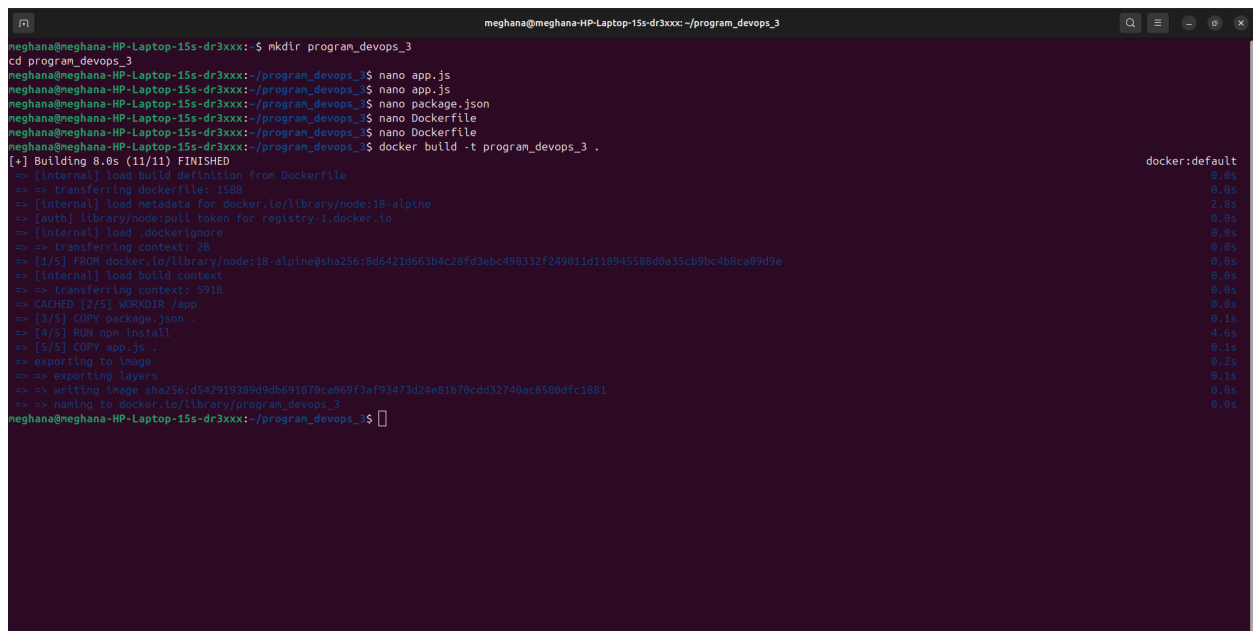
COPY app.js .

EXPOSE 4000

CMD ["npm", "start"]

[+] Read 12 lines
[+] Location
[+] Go To Line
[+] Undo
[+] Redo
[+] Set Mark
[+] Copy
[+] To Bracket
[+] Previous
[+] Back
[+] Next
[+] Forward
[+] Help
[+] Exit
[+] Write Out
[+] Read File
[+] Where Is
[+] Replace
[+] Cut
[+] Paste
[+] Execute
[+] Justify
```

Step 4 – Build Docker Image



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3
meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program_devops_3
cd program_devops_3
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$ nano package.json
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$ docker build -t program_devops_3 .
[+] Building 8.0s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> == transferring dockerfile: 158B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> == transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c20fd3ebc498332f249011d118945588d0a35cb9bc4b0ca89d9e
=> [internal] load build context
=> == transferring context: 591B
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package.json .
=> [4/5] RUN npm install
=> [5/5] COPY app.js .
=> exporting to image
=> == exporting layers
=> == writing image sha256:ds42919389d9db691870ca869f3af93473d24e81b70cdd32740ac8580dfc1881
=> == naming to docker.io/library/program_devops_3
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_3$
```

Step 5 – Run Docker Container

```
docker run -p 4000:4000 program_devops_3
```

Slide 8: Output Verification

Application URL:

<http://localhost:4000>

